Anonymity and Free Speech: How Political Context Shapes Wikipedia Editors

CPSC-298 Wikipedia Governance Research Project

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Abstract

Wikipedia aims to be a public encyclopedia, a place where people can share knowledge for the greater good. It works as a global digital public square for universal contributions of knowledge. While its intentions- and those of most contributors– are noble, a few bad actors can move such a resource from being a public square– one where the sharing of information is done respectfully for the pursuit of the truth– to a tool to spread propaganda, misinformation and defamation for their own personal gain. The unique open editing model allows anyone to contribute, but also offers an interesting outlook into trust and anonymity. Wikipedia's edits are tracked, and the Wikipedia API allows open access into the IP addresses of unregistered users, in part to prevent vandalism. While this helps in terms of moderating Wikipedia, it raises questions of how free speech plays a role, especially for editors who may live in countries under repressive political regimes. This research investigates the relationship between political context, free speech, and participation on Wikipedia. This informs and increases understanding on platform governance, user trust, and privacy; specifically how politics can change participation in digital public squares.

Keywords

Wikipedia, governance, [add your keywords here]

1 Introduction

The main concept that we will be investigating throughout this paper is speech. The way that speech is expressed and restricted is a significant factor into determining human behavior. It is equally important in understanding the health of public squares, whether they even exist, and who contributes to them. In this paper, we will first analyze and discuss this concept as a whole, and understand the correlation between free speech laws and the sharing of information. Once we have established this link, we will analyze whether or not a sense of surveillance has a chilling effect on the contribution of the uncensored sharing of information. At this point, we will look to Wikipedia for our data analysis, as it is the most commonly used example of a public square. Due to its decentralized, open nature, it is not linked to one government or entity that could take punitive measures for free speech. This model is also responsible for how truth is found on Wikipedia, with multiple potentially anonymous editors attempting to extract data from opinion, and come to an unbiased conclusion on different topics. This makes it useful for our analysis, as it is an online public square, one where contributions are documented. We will use Wikipedia as a microcosm for a public square, one with global users and try to understand how different freedom of speech laws affect contributions to these public squares. We do this by tracking where Wikipedia contributions are coming from via IP address, and breaking down contributions by country. We then take data from the V-Dem democracy index to quantify freedom of speech laws country by country.

This paper investigates the following research questions:

(1) Does surveillance have a chilling effect on speech in public forums? Is this dependent on enforcement or just the feeling of being surveilled? How does this relate to Wikipedia as a case study?
(2) Do freedom of speech laws have a chilling effect on the number of anonymous Wikipedia contributions?
(3) How do governance and administrative controls impact trust and anonymity?

The main contributions of this work are:

- Providing quantitative evidence to link free speech laws and Wikipedia edits.
- Show how governance mechanisms can impact participation and trust.
- Provide insights into balancing privacy and protection against vandalism in online public squares.

The rest of this paper is organized as follows. Section 2 reviews related work on Wikipedia governance. Section 3 describes our data and methods. Section 4 presents our findings. Section 5 discusses implications and limitations. Section 6 concludes and suggests future work.

2 Related Work

Wikipedia's unique open editing model offers an opportunity to research its governance in relation to politics and free speech laws.

2.1 Wikipedia Governance and Community Culture

Reagle's seminal work on Wikipedia culture [5] examines the collaborative practices and governance mechanisms that enable Wikipedia's success.

2.2 Wikipedia Trust and Reputation

de Laat studied how Wikipedia monitors edits in order to avoid vandalism, with both admins and autonomous bots [1]. Their paper states that Wikipedia experiences roughly 9000 vandalism edits daily, which is 7% of all edits. It elaborates on institutional trust, which in this context is how Wikipedia approaches trusting contributors. Wikipedia uses backgrounding trust, which requires monitoring reputation and uses corrective mechanisms in the background to enforce norms, with sanctions placed on users who deviate. Bots are used to highlight edits that are potentially vandalism, which raises the question of bias within the automated bots.

A paper on chilling effects showed that awareness of surveillance of IP address impacted Wikipedia usage, which is relevant because it means that potential editors/admins in more repressive countries will not participate to the full extent due to worries of surveillance [4]. The paper uses empirical backing to show the cost of spam control, with limits on edits, potentially especially for politically controversial pages.

2.3 Wikipedia and Political Context

The study done by Parra [3] uses a casual inference framework and showed that politicians who switched from left to right wing parties had a drop in sentiment on their Wikipedia articles, which could show systematic political bias. This means that even with Wikipedia's monitoring of vandalism there is bias, making it important for us to study the role of free speech and politics with Wikipedia users. An Oxford paper found that the majority of articles about places were edited by people living in the UK, USA, France, Germany, and Italy [2]. Digital connectivity is a contributing factor, along with high income levels.

2.4 Our Work in Context

The majority of prior existing similar research examines survey-based measures; we are using quantitative empirical behavioral data to examine user participation on Wikipedia. Our research is primarily focused on observable edits and behavior. Additionally, we focus on several different countries and create a bridge between quantitative edit evidence and free speech laws related to those countries. This contributes to further understanding how free speech laws can impact user behavior in digital public squares like Wikipedia.

3 Methodology

Using the Wikipedia API, we will track edits from frequently visited versions of Wikipedia for different countries for edits made by unregistered users. We will use the IP addresses of those unregistered users to link them to that editor's country. This will be used to compare edit frequency per country against benchmarks for free speech laws. Our benchmarks for free speech laws will be sourced from the V-Dem democracy index.

3.1 Data Collection

We used the Wikipedia API for frequently visited version of Wikipedia for different countries. We collected IP address information for edits by unregistered users.

3.2 Data Processing

We filtered out edits made by registered users, as we do not have access to their IP addresses. Unregistered users were linked to their respective country by IP address. Our data processing pipeline is available in our GitHub repository.[1]

3.3 Analysis Methods

[Describe your analytical approach. What techniques did you use? Why?]

3.4 Ethical Considerations

Optional: Include this if your project involves human subjects data, user behavior analysis, or if your institution requires ethics discussion. For basic Wikipedia article analysis, this may not be necessary - just ensure proper citation in your Data Collection section.

Example: All data consists of publicly available Wikipedia content accessed in compliance with Wikimedia's Terms of Use.

4 Results

[Brief paragraph introducing your main findings]

4.1 [Finding 1 - descriptive title]

[Present your first main finding]

4.2 [etc]

[Present your second main finding]

5 Discussion

5.1 Interpretation of Results

[What do your findings mean? How do they answer your research questions?]

5.2 Implications

[What are the broader implications of your work? For Wikipedia? For research? For practice?]

5.3 Limitations

[Be honest about limitations: data constraints, methodological issues, scope boundaries, etc.]

5.4 Future Work

[What questions remain? What should future research investigate?]

6 Conclusion

[Restate the problem you investigated]

[Summarize your approach and key findings]

[Emphasize your main contributions]

[End with a forward-looking statement about the importance of your work or future directions]

Acknowledgments

We thank …for ….

[1] https://github.com/kochenderferc/wikipedia-political-context

References

[1] Paul B de Laat. 2014. From open-source software to Wikipedia:'Backgrounding'trust by collective monitoring and reputation tracking. *Ethics and Information Technology* 16, 2 (2014), 157–169.

[2] University of Oxford. 2015. *Wikipedia world view 'shaped by editors in the West'*. https://www.ox.ac.uk/news/2015-09-15-wikipedia-world-view-shaped-editors-west

[3] Guillermo Parra. 2024. Righting the Writers: Assessing Bias in Wikipedia's Political Content-An Event Study and Sentiment Analysis. *Available at SSRN 5022797* (2024).

[4] Jonathon W Penney. 2016. Chilling effects: Online surveillance and Wikipedia use. *Berkeley Tech. LJ* 31 (2016), 117.

[5] Joseph Michael Reagle. 2010. *Good Faith Collaboration: The Culture of Wikipedia.* MIT Press.

A AI Usage Documentation

A.1 Literature Review

[Describe how you used AI agents for literature review. Reference your .prompt.md file.]

Example: We used an AI agent workflow (see the file literature-review.prompt.md) to systematically process research papers. The agent extracted summaries, methodology descriptions, and key findings from papers in our bibliography.

A.2 Data Analysis

[If you used AI for data analysis, code generation, or statistical work, document it here]

A.3 Writing Assistance

[Document any AI assistance in writing: brainstorming, editing, restructuring, etc.]

Example: We used Claude/ChatGPT to help with [specific task, e.g., "improving clarity of the abstract" or "suggesting visualizations for our data"].

A.4 Code Development

[If AI helped you write code for data collection or analysis, document it]

A.5 Verification

[How did you verify AI-generated content? What human oversight did you apply?]

All AI-generated content was reviewed, verified against primary sources, and edited by the human author(s). Factual claims were cross-checked with original papers and data.